

OpenClaw Active Interaction

OpenClaw Active Interaction

1. Course Content
 - Course Overview
 - Preparation
2. Creating a New Skill
3. Creating a Skill for Handling Special Situations
4. Case Demonstration
5. Extended Applications
 - 5.1 Equipment Fault Pre-warning and Handling
 - 5.2 Environmental Safety Monitoring
 - 5.3 Production Progress Tracking and Notifications
 - 5.4 Multi-Condition Composite Trigger Strategy

1. Course Content

Course Overview

- Based on "skills," you will write response strategies for specific scenarios, enabling OpenClaw to actively provide solutions when encountering unexpected situations.
- This section is a course on solution design tailored to specific application scenarios; therefore, you must first master the prerequisite courses: [OpenClaw Basics], [OpenClaw Interaction Methods], and [OpenClaw Skills Development].

Preparation

- First, complete the [OpenClaw Skills Development] course section to master the process of creating new skills.
- Launch the MicroROS chassis agent (if it is already running, do not launch it again).

```
sh start_agent.sh
```


3. Creating a Skill for Handling Special Situations

```
---
name: Production Line Material Shortage Handling Process
description: When performing inspection tasks, provides a preset handling process
for situations involving material shortages.
version: 1.0.1
---
```

```
## Production Workshop Colored Material Shortage Handling Plan
- Actively offers a solution to the user via a verbal inquiry; the corresponding
response strategy will be executed only after explicit confirmation from the
user.
## Option 1: Replenish from the raw material warehouse shelves
## Option 2: Proceed to the storage area and replenish using materials of the
same specifications but a different color
```

- After creation is complete, restart the OpenClaw gateway for the changes to take effect

```
openclaw gateway restart
```

4. Case Demonstration

[!NOTE]

- You may write the commands yourself according to your specific needs. The following serves as a case demonstration; you may choose any interaction method you prefer. Here, the WebChat interface is used as an example.

main
qwen3.6-plus · baillian

You are currently in the production workshop, checking the status of materials.

openclaw-control-ui 11:16

3 tools read, exec, SeeWhat

Assistant 11:16 114.2k 1300 1% ctx qwen3.6-plus

1 tool read

Tool output exec

Tool output SeeWhat

Tool 11:16

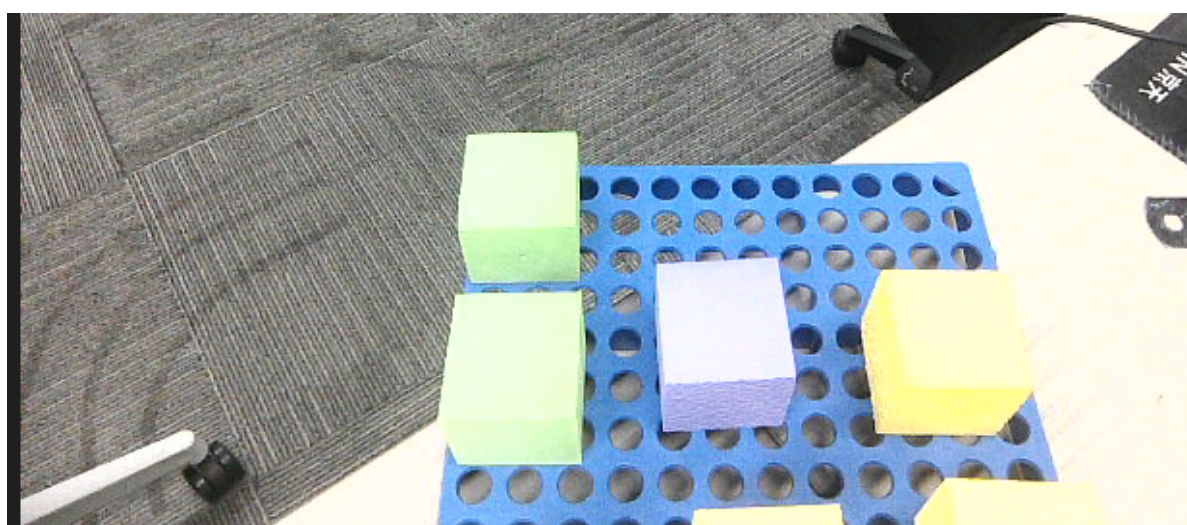
<tts>I'm at the production workshop, and I can see a blue pegboard with colorful cubes — one purple, three yellow, and three green blocks. Let me check the map and see if there are any specific tasks I should be handling here.</tts>
I can see the current workspace clearly:
Materials on the pegboard:

Color	Count
Purple	1
Yellow	3
Green	3

Total: 7 blocks — all neatly placed on the blue pegboard.
There's also a red object visible on the left edge (partially out of frame). Everything appears to be in order on the board.
Is there anything specific you'd like me to check or do with these materials? For example:

- Sort them into bins?
- Move them to another location?
- Count and verify inventory?

Message Assistant (Enter to send)



5. Extended Applications

OpenClaw's proactive interaction capabilities can be applied to a wide variety of scenarios. The following provides several reference examples for custom solutions:

5.1 Equipment Fault Pre-warning and Handling

```
---
name: Equipment Anomaly Handling Process
description: Proactively provides diagnostic and remedial suggestions when
abnormal equipment operating parameters are detected
version: 1.0.0
---
```

Proactive Equipment Anomaly Warning Solution

- Monitors equipment operating status in real-time; proactively alerts the user when anomalies are detected
- Provides tiered handling suggestions: notifications for minor anomalies, and recommendations for immediate shutdown in cases of severe anomalies
- Corresponding handling strategies are executed only after explicit confirmation by the user

Handling Option 1: Parameter Calibration

- Verify the accuracy of sensor readings
- Execute the equipment's self-diagnostic program
- Calibrate deviation parameters

Handling Option 2: Shutdown and Maintenance

- Immediately halt equipment operation
- Notify maintenance personnel to report to the site
- Log fault details for subsequent analysis

5.2 Environmental Safety Monitoring

```
---
name: Environmental Safety Monitoring Strategy
description: Proactively implements safety measures when environmental parameters
exceed safety thresholds
version: 1.0.0
---
```

Proactive Environmental Safety Monitoring Solution

- Continuously monitors environmental parameters such as temperature, humidity, and gas concentration levels
- - Immediately issue an alert to the user upon detection of a hazardous situation
- Provide emergency response protocols, requiring user confirmation before execution

Emergency Protocol 1: Ventilation and Air Exchange

- Activate the exhaust ventilation system
- Open doors and windows to facilitate airflow
- Continuously monitor air quality until normal levels are restored

Emergency Protocol 2: Emergency Evacuation

- Trigger the alarm system
- Guide personnel to evacuate in an orderly manner
- Contact professional emergency response agencies

5.3 Production Progress Tracking and Notifications

```
---
name: Proactive Production Progress Notifications
description: Periodically check the status of production tasks and proactively
notify users at key milestones
version: 1.0.0
---
```

Proactive Production Progress Tracking Strategy

- Periodically query the completion status of tasks on the production line
- Proactively notify users at key milestones (e.g., material shortages, schedule delays, task completion)
- Provide recommendations for next steps/actions

Notification Policies

- Materials Nearing Depletion: Provide a replenishment reminder 2 hours in advance
- Schedule Behind Plan: Analyze the root cause and offer recommendations for expediting work
- Task Successfully Completed: Generate a completion report and notify relevant personnel

5.4 Multi-Condition Composite Trigger Strategy

```
---
name: Intelligent Response to Composite Conditions
description: Combine multi-dimensional conditions—such as time, environment, and
equipment status—to provide intelligent, proactive services
version: 1.0.0
---
```

Multi-Dimensional Condition Response Strategy

- Comprehensively analyze data from various sensors alongside business logic
- Proactively trigger a response when a specific combination of conditions is met
- Provide personalized recommendations for handling the situation

Trigger Condition Examples

- Time Condition: During the night shift (22:00 – 06:00)
- Environmental Condition: Temperature exceeds 35°C AND humidity falls below 30%
- Equipment Condition: Continuous operation for more than 8 hours

Response Policies

- Proactively recommend equipment downtime for maintenance
- Alert staff members to potential environmental safety risks
- Automatically adjust equipment operating parameters to an energy-saving mode